DIRECTIONS *for questions 1 to 5:* Answer these questions on the basis of the information given below.

Eight persons, A through H, wanted to meet for dinner at a restaurant. The eight of them arrived at the restaurant at different times. Except for the person who was the last to arrive at the restaurant, each person waited outside the restaurant until the last person arrived. As soon as the last person arrived at the restaurant, they went into the restaurant and were seated around a circular table in eight equally spaced chairs.

Further, the times at which the eight persons arrived at the restaurant are 7:10 PM, 7:20 PM, 7:45 PM, 8:10 PM, 8:45 PM, 9:10 PM, 9:25 PM and 9:35 PM.

It is also known that

i. A, who waited longer than B, was sitting opposite the person who was the third to arrive at the restaurant.

ii. the person who was sitting to the left of C waited for sixty minutes more than C did.

iii. F and G, who were sitting adjacent to each other, arrived one immediately after the other, in the same order.

iv. H, who was not sitting adjacent to F, was sitting adjacent to the person who arrived at 9:25 PM.

v. the number of minutes that D waited was twice the number of minutes that the person sitting to the right of D waited.

vi. for one pair of persons sitting opposite each other, the total time that the two of them waited was more than 240 minutes and for one other pair of persons sitting opposite each other, the total time that the two of them waited was exactly 110 minutes.

Q1. **DIRECTIONS** *for question 1:* Select the correct alternative from the given choices.
Who was the first person to arrive at the restaurant?

- a) **B**
- b) **E**
- c) **F**
- d) **Cannot be determined**

Q2. DIRECTIONS *for question 2:* Type in your answer in the input box provided below the question.

How many minutes did the person sitting opposite B wait?

Q3. DIRECTIONS *for questions 3 to 5:* Select the correct alternative from the given choices.

Who is sitting opposite H?

a) **C**

b) **D**

c) **A**

d) **E**

Q4. DIRECTIONS *for questions 3 to 5:* Select the correct alternative from the given choices.

Among the persons who arrived after 8:00 PM, how many pairs of persons are sitting opposite each other?

- a) **0**
- b) **1**
- c) **3**
- d) **2**

Q5. DIRECTIONS *for questions 3 to 5:* Select the correct alternative from the given choices.

How many persons arrived before F?

- a) **5**
- b) **6**
- c) **4**
- d) **3**

DIRECTIONS *for questions 6 to 10:* Answer these questions on the basis of the information given below.

Pavan purchased a new car at the beginning of January 2021. The car came with a total of five tires – four tires fixed to the car and one tire as a spare. The positions of the four tires fixed to the car are Front Left, Front Right, Back Left and Back Right. The total distance that any tire travels is referred to as the mileage of that tire. When buying the car, the technician informed Pavan that whenever the mileage of any tire reaches 10000 km, that tire must be replaced with a new one. The technician also advised that every month, Pavan should remove one of the four tires fixed to the car and swap it with the spare tire, so that the mileages of all the tires remain as equal as possible.

From the beginning of January 2021 to the end of June 2022, Pavan travelled a distance of exactly 2000 km each month. However, he did not swap the tires every month as advised but only did so at the beginning of certain months.

The table below provides, for the given period, the months at the beginning of which he swapped the tires and the position of the tire which he swapped. When swapping a tire, the spare tire is placed in the position of the tire being swapped and the swapped tire becomes the spare tire. He did not change the positions of any other tires at any other point of time. Further, he kept a track of the mileage of each tire and discarded any tire as soon as its mileage reached 10000 km and attached a new tire in its position. If any tire reached a mileage of 10000 km at the end of a month, he discarded it and purchased a new tire before the beginning of the next month.

Month	Position of the Swapped Tire
March 2021	Front Left
July 2021	Back Right
September 2021	Front Left
December 2021	Back Left
February 2022	Front Right
May 2022	Back Right

Q6. DIRECTIONS *for questions 6 and 7:* Select the correct alternative from the given choices.
How many tires did Pavan purchase from the beginning of January 2021 to the beginning of May 2022 (not counting the five tires that came with the new car)?

- a) **10**
- b) **11**
- c) **12**
- d) **13**

Q7. DIRECTIONS *for questions 6 and 7:* Select the correct alternative from the given choices.
What is the mileage of the tire which was the spare tire on 20th May 2022?

- a) **2000 km**
- b) **4000 km**
- c) **8000 km**
- d) **0 km**

Q8. DIRECTIONS *for question 8:* Type in your answer in the input box provided below the question.

What is the sum of the mileages (in km) of the five tires in Pavan's car at the beginning of November 2021?

Q9. DIRECTIONS *for question 9:* Select the correct alternative from the given choices.
At the beginning of how many months during the period from the beginning of January 2021 to the end of June 2022 did each of the five tires in Pavan's car have a mileage of more than 3000 km?

- a) **3**
- b) **1**
- c) **4**
- d) **5**

Q10. DIRECTIONS *for question 10:* Type in your answer in the input box provided below the question.

At the beginning of how many months during the period from the beginning of January 2021 to the end of June 2022 was the mileage of the tire in the Front Right position more than that of the tire in the Back Right position?

DIRECTIONS *for questions 11 to 15:* Answer these questions on the basis of the information given below.

Five traders – Lalit, Manoj, Naveen, Omar and Piyush – purchased rice at different prices per kg among ₹45, ₹68, ₹73, ₹77 and ₹85, not necessarily in the same order. Further, each trader sold the purchased rice at a different price per kg among ₹78, ₹84, ₹98, ₹106 and ₹124, not necessarily in the same order. It is also known that the profit per kg made by each trader is distinct, and no trader made a profit of more than ₹40 per kg in selling the rice. The following information is also known:

i. Lalit, who made a profit greater than ₹35 per kg, did not purchase the rice at the highest price.

ii. Manoj, who did not purchase the rice at the lowest price, purchased the rice for less than ₹75 per kg.

iii. The least profit that anyone made was greater than ₹10 per kg.

iv. The profit per kg made by Omar was greater than that made by Piyush but neither of them sold the rice for more than ₹100 per kg.

Q11. **DIRECTIONS** *for question 11:* Type in your answer in the input box provided below the question.
How many traders made a profit greater than ₹30 per kg?

Q12. DIRECTIONS *for questions 12 to 14:* Select the correct alternative from the given choices.
Who made the highest profit per kg?

- a) **Manoj**
- b) **Lalit**
- c) **Naveen**
- d) **Omar** ✗ Your answer is incorrect

Q13. **DIRECTIONS** *for questions 12 to 14:* Select the correct alternative from the given choices.
Which of the following statements are true?

I. The trader who purchased the rice at the highest price made the highest profit per kg.

II. The trader who sold the rice at the highest price made the highest profit per kg.

III. The trader who sold the rice at the lowest price made the lowest profit per kg.

IV. The trader who purchased the rice at the lowest price made the lowest profit per kg.

- a) **I and IV**
- b) **II and III**
- c) **I and II**
- d) **I, II and IV**

Q14. **DIRECTIONS** *for questions 12 to 14:* Select the correct alternative from the given choices.
What is the average of the profits per kg made by the five traders?

- a) **₹25.60**
- b) **₹31.40**
- c) **₹29.60**
- d) **₹28.40**

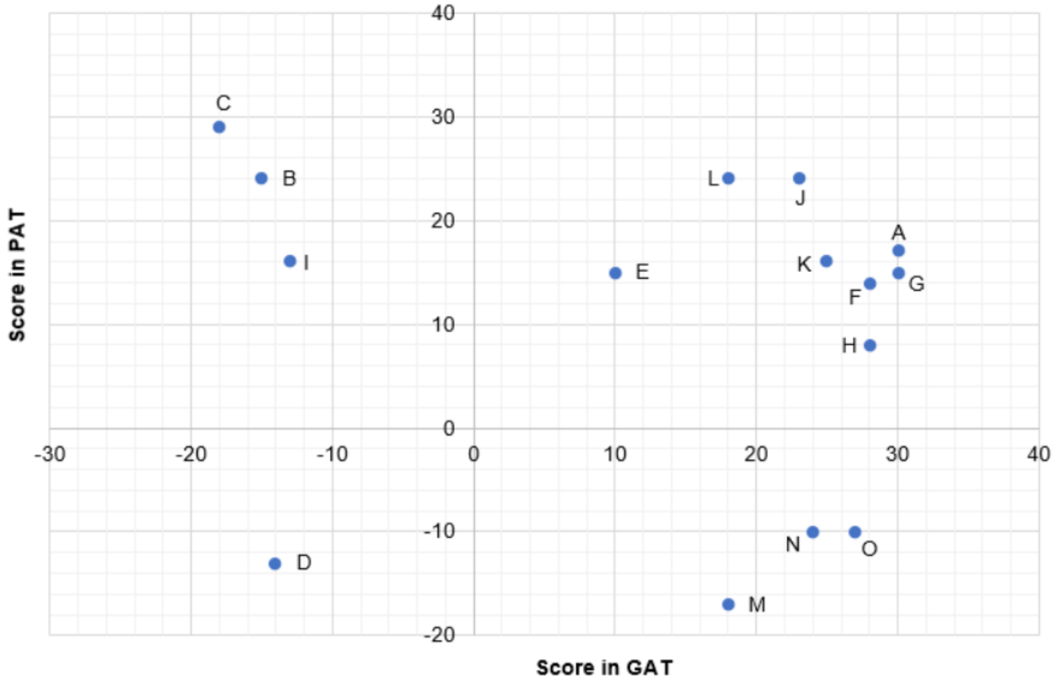
Q15. DIRECTIONS *for question 15:* Type in your answer in the input box provided below the question.

What is the profit per kg made by Piyush?

DIRECTIONS *for questions 16 to 20:* Answer these questions on the basis of the information given below.

Fifteen students, A through O, attempted two exams – GAT and PAT. The score of any student in any exam was an integer. Further, the 15 students belong to five different colleges – C1 through C5. The number of students from C1 through C5 are 2, 4, 3, 4 and 2, respectively.

The graph below provides the scores of the 15 students in the two exams, with the PAT scores along the vertical axis and the GAT scores along the horizontal axis.



It is also known that

i. no two students from the same college had the same score in either GAT or PAT.

ii. the score in GAT of any student from C1 was less than that of any student from C2, while the score in PAT of any student from C1 was not less than that of any student from any of C2 or C3 or C4.

iii. the score in PAT of any student from C5 was less than that of any student from any of C1 or C2 or C4.

iv. no student from C4 scored less than 14 in either of the two tests, while every student from C3 scored less than –5 in PAT.

Q16. DIRECTIONS *for questions 16 to 20:* Select the correct alternative from the given choices.
Which college does A belong to?

- a) **C2**
- b) **C4**
- c) **C3**
- d) **Either C2 or C4**

Q17. DIRECTIONS *for questions 16 to 20:* Select the correct alternative from the given choices.
What is the sum of the scores in PAT of all the students from C2?

- a) **70**
- b) **72**
- c) **76**
- d) **Either 70 or 72**

Q18. DIRECTIONS *for questions 16 to 20:* Select the correct alternative from the given choices.
Which of the following options represents the exhaustive list of colleges to which N can belong?

- a) **C2, C3, C5**
- b) **C5**
- c) **C2, C3, C4**
- d) **C3, C5**

Q19. DIRECTIONS *for questions 16 to 20:* Select the correct alternative from the given choices.
What BEST can be said about the difference between the sum of the scores in PAT of all the students in C4 and the sum of the scores in GAT of all the students in C4?

- a) **It lies between 29 and 38.**
- b) **It is either 32 or 37.**
- c) **It lies between 25 and 30.**
- d) **It is either 27 or 28.**

Q20. DIRECTIONS *for questions 16 to 20:* Select the correct alternative from the given choices.
If the sum of the scores in PAT of the students from C1 is more than the sum of the scores in GAT of the students from C5, to which college does O belong?

a) **C5**

b) **C4**

c) **C3**

d) **Either C3 or C4**